$\phi = 0.05 \sin 377t$ wb

If there are 100 turns on the core, what voltage is produced at the terminal of the coil.

Solution: Given that, $\phi = 0.05 \sin 377t$ wb

we know that, $\mathcal{E} = N \frac{d\phi}{dt}$

$$\therefore \frac{d}{dt}(\phi) = \frac{d}{dt}(0.05 \sin 377t)$$
$$= 0.05 \cos 377t \cdot \frac{d}{dt}(377t)$$
$$= 0.05 \cos 377t \cdot 377$$
$$= (0.05 \times 377) \cos 377t$$
$$= 18.85 \cos 377t$$

$$\therefore \mathcal{E} = N \frac{d\phi}{dt}$$
$$= 100 \times 18.85 \cos 377t$$
$$= 1885 \cos 377t \text{ V}$$

(Ans)

A coil that has 700 turns develops an average induced voltage of 50 V. What must be the change in the magnetic flux occur to produce such a voltage if the time interval for his changes is 0.7 seconds.

Solution: Given that, $N = 700$
$t = 0.7$ seconds
$\mathcal{E} = 50$ V

we know that, $\mathcal{E} = N \frac{\phi}{t}$

$$\Rightarrow \mathcal{E} = \frac{N\phi}{t}$$
$$\Rightarrow N\phi = \mathcal{E}t$$
$$\Rightarrow \phi = \frac{\mathcal{E}t}{N} = \frac{50 \times 0.7}{700} = 0.05 \text{ wb}$$

(Ans)

The magnetic flux linked with a coil having 250 turns is changed from 1.4 wb to 2 wb in 0.45 sec. calculate the induced e.m.f.

Solution: Given that, $N = 250$

$\phi_1 = 1.4$ wb

$\phi_2 = 2$ wb

$t = 0.45$ sec

we know that, Induced emf, $\varepsilon = N \frac{\phi}{t}$

$= N \times \frac{\phi_2 - \phi_1}{t}$

$= 250 \times \frac{2 - 1.4}{0.45}$

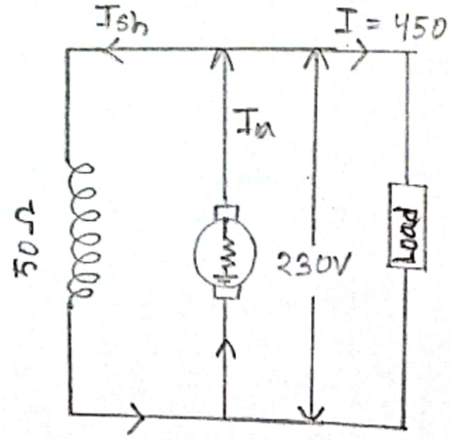
$= 333.33$ V

(Ans)

26.3 A shunt generator deliver 450A at 230V and the resistance of the shunt field and armature are 50Ω and 0.03Ω respectively. Calculate the generator e.m.f.

Solution: Current through shunt field winding is, $I_{sh} = \frac{230}{50}$ $[V = IR, I = \frac{V}{R}]$

$= 4.6A$



Load current, $I = 450A$

$\therefore$ Armature current, $I_a = I + I_{sh}$

$= 450 + 4.6$

$= 454.6$ A

Armature voltage drop, $V = I_a R_a$

$= 454.6 \times 0.03$ V

$= 13.638$ V

$\therefore$ e.m.f generated in armature, $E_g = I_A R_A + V_t$

$= \cancel{230}\ 13.6 + 230$

$= 243.6\,V$

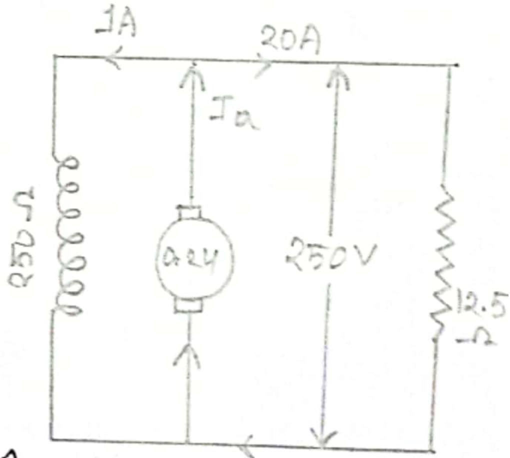
(Ans)

26.11 (a) An 8 pole d.c shunt generator with 778 wave-connected armature conductors and running at 500 r.p.m. supplies a load of 12.5 Ω resistance at terminal voltage of 50V. The armature resistance is 0.24 Ω and the field resistance is 250 Ω. Find armature current induced e.m.f and flux per pole.

Solution: Given that,

$V = 250V$

$R = 12.5\,\Omega$



Load current, $I = \frac{V}{R} = \frac{250}{12.5}$

$= 20A$

Shunt current, $= \frac{V}{R} = \frac{250\,V}{250\,\Omega} = 1\,A$

Armature current $= (20+1)\,A$

$= 21\,A$

Induced e.m.f, $E_g = I_A R_A + V_t$

$= (21 \times 0.24 + 250)$

$= 255.04\,V$

Now, $E_g = \frac{\Phi 2NP}{60A}$

$\Rightarrow E_g\,60A = \Phi 2NP$

$\Rightarrow \Phi = \frac{E_g\,60A}{2NP}$

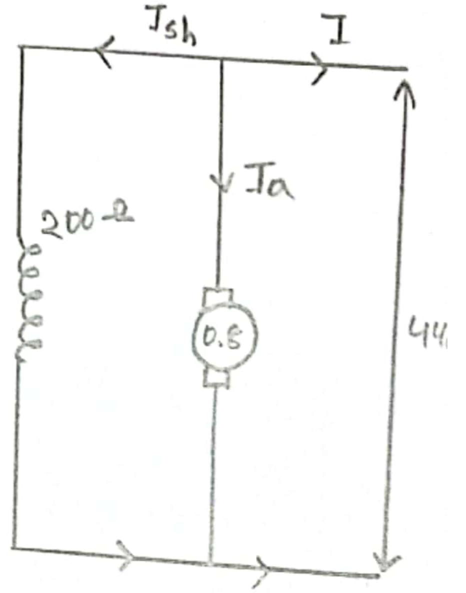
$= \frac{255.04 \times 60 \times 2}{778 \times 500 \times 8}$

$= 0.00983\ wb$

$= 9.83\ m\,wb$ (Ans)

22.3 A 440V shunt motor has armature resistance of 0.8Ω and field resistance of 200Ω. Determine the back e.m.f. when giving an output of 7.46kw at 85 percent efficiency.

Solution: Given that,



$$\text{Motor input unit} = 7.46\,kW$$
$$= 7.46\times10^3\ W$$
$$= \frac{7.46\times10^3}{85\%}\ W$$
$$= \frac{7460}{0.85}\ W$$
$$= 8776.47\ W$$

Motor input current,
$$I = \frac{P}{V} = \frac{8776.47}{440} = 19.95\,A$$

Shunt current,
$$I_{sh} = \frac{440}{200} = 2.2A$$

By using KCL, $I = I_A + I_{sh}$

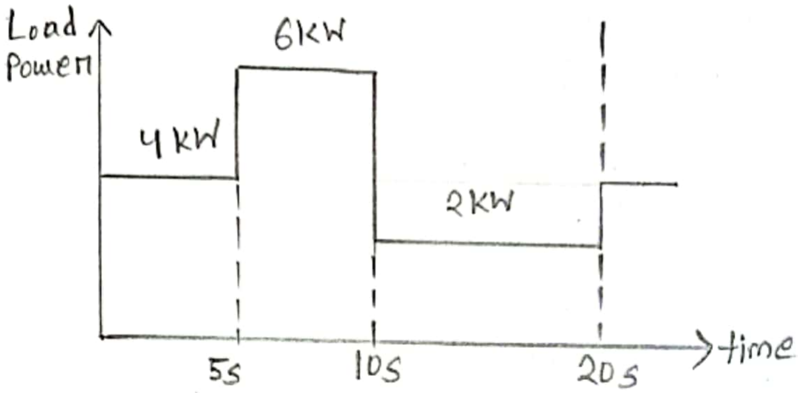
$$\Rightarrow I_A = I - I_{sh}$$
$$= 19.95 - 2.2$$
$$= 17.75\ A$$

By using KVL, $V_t + I_A R_A - E_b = 0$

$$\Rightarrow E_b = V_t - I_a R_a$$
$$= 440 - (17.75\times0.8)$$
$$= 440 - 14.2$$
$$= 425.8\ V$$

(Ans)

⊡ The following load torque profile was measured for a constant speed load. Choose an appropriately sized motor to drive this load assuming a service factor of 20%. You do not need to consider starting performance.



Solution: Peak load power requirement = 6 KW

Peak load power must be less than Rated Motor Power × service factor

load torque > rated motor torque $\left(1+\frac{\text{service factor}}{100\%}\right)$

$\Rightarrow 6\,KW < P_{motor} \times 1.2$

$\Rightarrow P_{motor} > 6/1.2$

$= 5\,KW$

RMS load power $= \sqrt{\dfrac{4^2\times5+6^2\times5+2^2\times10}{5+5+10}}$

$= 4.16\,KW$

If we choose $P_{motor} = 5\,KW$ then RMS load power $= \dfrac{4.16}{5}\times100\%$

$= 83\%$ of P_{motor} which is acceptable.

- The RMS load power should ideally be between 75% and 100% of rated motor power requirement.

∴ choose a 5KW motor

(Ans)

32.1 The maximum flux density in the core of a 250/3000-Volts, 50-Hz. Single phase transformer is 1.2 wb/m². If the e.m.f per turn is 8 Volt, determine,

(i) Primary and Secondary turns (ii) Area of the core

Solution: (i)

$$E_1 = N_1 \times \text{e.m.f induced/turn}$$

$$\Rightarrow N_1 = \frac{E_1}{\text{e.m.f. induced per turn}}$$

$$= \frac{250}{8}$$

$$= 32$$

$$E_2 = N_2 \times \text{e.m.f induced/turn}$$

$$\Rightarrow N_2 = \frac{E_2}{\text{e.m.f induced per turn}}$$

$$= \frac{3000}{8}$$

$$= 375$$

(Ans)

(ii) we may use,

$$E_1 = -4.44 f N_2 B_m A$$

$$\text{or, } A = \frac{E_1}{4.44 f N_2 B_m}$$

$$= \frac{250}{4.44 \times 50 \times 32 \times 1.2}$$

$$= 0.03\, m^3$$

(Ans)

32.2 ~~A single phase~~ The core of a 100-kVA, 11000/550 V, 50 Hz, 1-Ph, core type transformer has a cross section of 20 cm × 20 cm. Find (i) the number of H.V and L.V turns per phase and (ii) the e.m.f. per turn if the maximum core density is not to ~~su~~ exceed 1.3 tesla. Assume a stacking factor of 0.9. What will happen if its primary voltage is increased by 10% on no-load?

Solution: Given that, $B_m = 1.3T$ $A = (0.2\times0.2)\times0.9 = 0.036 m^2$

~~⇒ 11000~~ ⇒ .

(i) $E_1 = 4.44 f N_1 B_m A$

or, $N_1 = \frac{E_1}{4.44 f B_m A}$

$= \frac{11000}{4.44\times50\times0.036\times1.3}$

$= 1058.75 \approx 1059$

$E_2 = 4.44 f N_2 B_m A$

$\Rightarrow N_2 = \frac{E_2}{4.44 f B_m A} = \frac{550}{4.44\times50\times0.036\times1.3}$

$= 52.95 \approx 53$

(ii) Given, $E_1 = 11000$

From (i) we get, $N_1 = 1059$

$\therefore$ emf/turn $= \frac{E_1}{N_1} = \frac{11000}{1059}$

$= 10.4 V$

Or, emf/turn $= \frac{E_2}{N_2} = \frac{550}{53}$

$= 10.4 V.$

32.3 A single phase transformer has 400 primary and 1000 secondary turns. The net cross-sectional area of the core is 60 cm². If the primary winding be connected to a 50-Hz supply at 520V. calculate (i) the peak value of flux density in the core (ii) the voltage induced in the secondary winding.

Solution: (i) Given that, $E_1 = 520V$
$f = 50$ Hz
$N_1 = 400$
$N_2 = 1000$

we know, ~~$E_1 = 520V$~~ $E_1 = 4.44 f N_1 B_m A$
~~$f = 50$ Hz~~ $\therefore$ Flux density, $B_m = \dfrac{E_1}{4.44 \times f \times N_1 \times A}$
~~$N_1 = 400$~~
~~$N_2 = 1000$~~

$$= \frac{520}{4.44 \times 50 \times 400 \times 60 \times 10^{-4}}$$

$$= 0.976 \text{ wb/m}^2$$

(ii) $k = \dfrac{N_2}{N_1} = \dfrac{50}{500} = \dfrac{1}{10} = 2.5$

$E_2 = kE_1$
$= 2.5 \times 520$
$= 1300V$

(Ans)

32.4 A 25 kVA transformer has 500 turns on the primary and 50 turns on the secondary winding. The primary is connected to 3000 V, 50 Hz supply. Find the full-load primary and secondary currents, the secondary e.m.f. and the maximum flux in the core. Neglect leakage drops and no-load primary current.

Solution: Given that, $N_1 = 500$

$N_2 = 50$

$P = 25\,kVA$

$= 25000\ VA$

$E_1 = 3000V$

$f = 50\ Hz$

$\therefore K = \frac{N_2}{N_1} = \frac{50}{500} = 0.1$

Now, full-load, $I_1 = \frac{P}{V}$

$= \frac{25000}{3000}$

$= 8.33\ A$

Full-load $I_2 = \frac{I_1}{k}$

$= \frac{8.33}{0.1}$

$= 83.3\ A$

$\therefore$ e.m.f per turn on primary side $= \frac{3000}{500}$

$= 6V$

Secondary e.m.f $= 6 \times 50$

$= 300V$

$\left[\text{OR, } \left(E_2 = kE_1 = 0.1 \times 3000 = 300V \right) \right]$

Again,

$E_1 = 4.44 f N_1 \Phi_m$

or, $\Phi_m = \frac{E_1}{4.44 f N_1}$

$= \frac{3000}{4.44 \times 50 \times 500}$

$= 0.027\ wb$

$= 0.027 \times 1000\ mwb$

$= 27\ mwb.$ (Ans)

32.9 (a) A 2,200/200 V transformer draws a no-load primary current of 0.6A and absorbs 400 watts. Find the magnetising and iron loss currents.

(b) A 2,200/250 V transformer takes 0.5A at a P.f of 0.3 on open circuit. Find magnetising and working components of no-load primary current.

Solution: Given that,

(a)

No-load input in watts 400 watts

Primary voltage 2200V

$$\therefore \text{Iron loss current} = \frac{\text{no-load Input in watts}}{\text{Primary voltage}}$$

$$= \frac{400}{2200}$$

$$= 0.182\ A$$

Now, $I_0^2 = I_W^2 + I_\mu^2 \Rightarrow I_\mu = \sqrt{I_0^2 - I_\omega^2}$

$\therefore$ Magnetising component, $I_\mu = \sqrt{(0.6)^2 - (0.182)^2}$

$$= 0.572\ A$$

(Ans)

(b) Given that,

$\cos\phi_0 = 0.3$

~~From~~ $I_0 = 0.5\ A$

$$\therefore I_W = I_0 \cos\phi_0$$

$$= 0.5 \times 0.3$$

$$= 0.15\ A$$

$$\therefore I_\mu = \sqrt{I_0^2 - I_\omega^2}$$

$$= \sqrt{(0.5)^2 - (0.15)^2}$$

$$= 0.476\ A$$ (Ans)

32.15 A 30KVA, 2400/120V, 50-Hz transformer has a high voltage winding resistance of 0.1Ω and a leakage ~~resistance~~ reactance of 0.22Ω Ω. The low voltage winding resistance is 0.035Ω and the leakage ~~resistance~~ reactance is 0.012Ω. Find the equivalent winding resistance, reactance and impedance referred to the (i) high voltage side and (ii) the low voltage side.

Solution: Given that,

$$E_1 = 2400\text{ V}$$
$$E_2 = 120\text{ V}$$
$$f = 50\text{ Hz}$$
$$P = 30\text{KVA} = 30000\text{VA}$$
$$R_1 = 0.1\,\Omega$$
$$R_2 = 0.035\,\Omega$$
$$X_1 = 0.22\,\Omega$$
$$X_2 = 0.012\,\Omega$$

$$\therefore k = \frac{120}{2400} = \frac{1}{20} = 0.05$$

(i) Here, high voltage side in primary side, values are referred to primary side,

$$R_{01} = R_1 + R_2' = R_1 + \frac{R_2}{k^2}$$
$$= 0.1 + \frac{0.035}{(0.05)^2} = 14.1\,\Omega$$

$$X_{01} = X_1 + X_2' = X_1 + \frac{X_2}{k^2}$$
$$= 0.22 + \frac{0.012}{(0.05)^2} = 5.02\,\Omega$$

$$Z_{01} = \sqrt{R_{01}^2 + X_{01}^2}$$
$$= \sqrt{(14.1)^2 + (5.02)^2}$$
$$= 14.96\,\Omega$$

(Ans)

(iii) Here, low voltage side is secondary side.

$$R_{02} = R_2 + R_1' = R_2 + R_1 k^2 = 0.035 + 0.1 \times (0.05)^2$$
$$= 0.03525\,\Omega$$

$$X_{02} = X_2 + X_1' = X_2 + X_1 k^2 = 0.012 + (0.22 \times 0.05^2)$$
$$= 0.01255\,\Omega$$

$$\therefore Z_{02} = \sqrt{R_{02}^2 + X_{02}^2}$$
$$= \sqrt{(0.03525)^2 + (0.01255)^2}$$
$$= 0.0374\,\Omega$$

(Ans)

32.16 A 50 KVA, 4400/220 V transformer has $R_1 = 3.45\,\Omega$, $R_2 = 0.009\,\Omega$. The values of reactances are $X_1 = 5.2\,\Omega$ and $X_2 = 0.015\,\Omega$. Calculate for the transformer (i) equivalent resistance as referred to primary (ii) equivalent resistance as referred to secondary (iii) equivalent reactance as referred to both primary and secondary (iv) equivalent impedance as referred to both primary and secondary, (v) total cu loss, first using individual resistances of the two windings and secondly, using equivalent resistance as referred to each side.

Solution: Full load primary current, $I_1 = \frac{50000}{4400} = 11.36\text{ A}$

Full load secondary current, $I_2 = \frac{50000}{220} = 227.27\text{ A}$

Given that, $R_1 = 3.45\,\Omega$ $\quad X_1 = 5.2\,\Omega$

$R_2 = 0.009\,\Omega$ $\quad X_2 = 0.015\,\Omega$

$$\therefore k = \frac{E_2}{E_1} = \frac{220}{4400} = 0.05.$$

(i) $R_{01} = R_1 + R_2' = R_1 + \frac{R_2}{K^2} = 3.45 + \frac{0.009}{(0.05)^2}$

$= 7.05\ \Omega$

(ii) $R_{02} = R_2 + R_1' = R_2 + R_1 K^2 = 0.009 + 3.45 \times (0.05)^2$

$= 0.0176\ \Omega$

(iii) $X_{01} = X_1 + X_2' = X_1 + \frac{X_2}{K^2} = 5.2 + \frac{0.015}{(0.05)^2} = 11.2\ \Omega$

$X_{02} = X_2 + X_1' = X_2 + K^2 X_1 = 0.015 + 5.2 \times (0.05)^2$

$= 0.028\ \Omega$

(iv) $Z_{01} = \sqrt{(R_{01})^2 + (X_{01})^2} = \sqrt{(7.05)^2 + (11.2)^2} = 13.234\ \Omega$

$Z_{02} = \sqrt{(R_{02})^2 + (X_{02})^2} = \sqrt{(0.0176)^2 + (0.028)^2} = 0.03311\ \Omega$

(v) Cu loss $= I_1^2 R_1 + I_2^2 R_2 = (11.36)^2 \times 3.45 + (227.27)^2 \times 0.009$

$= 910.08\ W$

Also, Cu loss $= I_1^2 R_{01} = (11.36)^2 \times 7.05 = 909.79 \approx 910\ W$

Cu loss $= I_2^2 R_{02} = (227.27)^2 \times 0.0176 = 909.06 \approx 910\ W$

(Ans)

32.21 A 230/460 V transform has a primary resistance of 0.2 Ω and reactance of 0.5 Ω and the corresponding values for the secondary are 0.75 Ω and 1.8 Ω respectively. Find the secondary terminal voltage when supplying 10A at 0.8 power factor lagging.

Solution: Given that, $E_1 = 230V$

$E_2 = 460V$

$R_1 = 0.2\ \Omega$

$R_2 = \cancel{0.7}\ 0.75\ \Omega$

$X_1 = 0.5\ \Omega$

$X_2 = 1.8\ \Omega$

$k = \frac{E_2}{E_1} = \frac{460}{230} = 2$

$R_{02} = R_2 + R_1 k^2 = 0.75 + 0.2 / (2)^2 = 1.55\,\Omega$

$X_{02} = X_2 + X_1 k^2 = 1.8 + \frac{0.5}{(2)^2} = 3.8\,\Omega$

Voltage drop $= (I_2 R_{02} \cos\phi + I_2 X_{02} \sin\phi)$

$= I_2 (R_{02} \cos\phi + X_{02} \sin\phi)$

$= 10 (1.55 \times 0.8 + 3.8 \times 0.6)$

$= 35.2V$

$\therefore$ secondary terminal voltage $= 460 - 35.2$

$= 424.8\,V$

(Ans)

32.22 Calculate the regulation of a transformer in which the percentage resistance drops is 1.0% and percentage reactance drop is 5.0% when the power factor is (a) 0.8 lagging (b) unity and (c) 0.8 leading.

Solution: (a) Power factor $= \cos\phi = 0.8$ lag

$\mu = V_r \cos\phi + V_x \sin\phi$

$= 1 \times \cos\phi + 5 \times \sin\phi$

$= 1 \times 0.8 + 5 \times 0.6$

$= 3.8\%$

(b) Power factor $= \cos\phi = 1$

$\mu = V_r \cos\phi + V_x \sin\phi$

$= 1 \times 1 + 5 \times 0 = 1\%$

(c) Power factor $= \cos\phi = 0.8$ leading

$\mu = V_r \cos\phi - V_x \sin\phi$

$= 1 \times 0.8 - 5 \times 0.6$

$= -2.2\%$ (Ans)

32.27 In no-load test of singl- phase transformer, the following test data were obtained:

Primary voltage : 220V ; secondary voltage: 110V;
Primary current: 0.5 A ; secondary current : 30W
Find the following: (I) The turns ratio
(II) The magnetising components of no-load current
(III) its working (or loss) component
(IV) the iron loss

Resistance of the primary winding = 0.6 ohm
Draw the no-load phasor diagram to scale

Solution: (I) $\frac{N_1}{N_2} = \frac{E_1}{E_2} = \frac{220}{110} = 2$

(II) we know, $W = V_0 I_0 \cos\phi_0$.

$$\text{or, } \cos\phi_0 = \frac{W}{V_0 I_0} = \frac{30}{220 \times 0.5} = 0.273$$

$$\therefore \sin^2\phi + \cos^2\phi = 1$$

$$\therefore \sin\phi = \sqrt{1-\cos^2\phi} = \sqrt{1-(0.273)^2} = 0.962$$

$$I_\mu = I_0 \sin\phi_0 = 0.5 \times 0.962$$
$$= 0.48\text{ A}$$

(III) $I_w = I_0 \cos\phi_0 = 0.5 \times 0.273 = 0.1365\text{ A}$

(IV) Primary cu loss $= I_0^2 R_1$
$= (0.5)^2 \times 0.6$
$= 0.15\text{ W}$

Iron loss $= 30 - 0.15$
$= 29.85\text{ W}$

(Ans)

32.28 A 5kVA 200/1000V, 50 Hz single phase transformer gave the following test results:

O.C Test (L.V. Side): 200V, 1.2A, 90W

① Calculate the parameters of the equivalent circuit referred to the L.V. side.

Solution: Shunt branch parameters from O.C test (LV side):

$R_0 = \frac{V^2}{P_0}$

$= \frac{(200)^2}{90}$

$= 444.44\ \Omega$

Here,

$V = 200V$

$I_0 = 1.2A$

$P_0 = 90W$

$I_W = \frac{V_0}{R_0} = \frac{200}{444.44}$

$= 0.45$ amp

$I_W = \sqrt{I_0^2 - I_w^2}$

$= \sqrt{(1.2)^2 - (0.45)^2}$

$= 1.11$ amp

$X_m = \frac{V}{I_\mu}$

$= \frac{200}{1.11}$

$= 180.18\ \Omega$

(Ans)

32.29 In a transformer, the core loss is found to be 52W at 40 Hz and 90W at 60Hz measured at same peak flux density. Compute the hysteresis and eddy current losses at 50 Hz.

Solution: Since the flux density is the same in both cases, we can use the relation,

Total core loss, $w_i = Af + Bf^2$

or, $\frac{w_i}{f} = A + Bf$

For 1st part, $\frac{52}{40} = A + 40B$

or, $A = \frac{52}{40} - 40B$ —(1)

For 2nd part, $\frac{90}{60} = A + 60B$

or, $\frac{90}{60} = \frac{52}{40} - 40B + 60B$ [From(1)]

or, $20B = \frac{90}{60} - \frac{52}{40} = \frac{1}{5}$

or, $B = 0.01$ And $A = \frac{52}{40} - 40 \times 0.01 = 0.9$

At 50 Hz, the two losses are,

$w_h = Af = 0.9 \times 50 = 45W$

$w_e = Bf^2 = 0.01 \times (50)^2 = 25W$

32.32 When a transformer is connected to a 1000 V, 50 Hz supply the core loss is 1000W of which 650 is hysterisis and 350 is eddy current loss. If the applied voltage is raised to 2000V and the frequency to 100 Hz. Find the new core losses.

Solution: Here, flux density unchanged,

with 1000V and 500 Hz

$w_h = Af = 50A$

or, $650 = 50A$

$\therefore A = 13$

$w_e = Bf^2 = 2500B$

or, $350 = 2500B$

$\therefore B = 0.14$

with 2000V and 100 Hz

$$W_h = Af = 13 \times 100 = 1300\,W$$

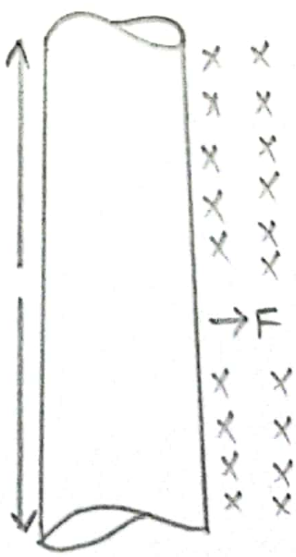
$$W_e = Bf^2 = 0.14 \times (100)^2 = 1400\,W$$

$\therefore$ New core loss $= 1300 + 1400$

$= 2700\,W$

(Ans)

1.7 Figure shows a wire carrying a current in the presence of a magnetic field. The magnetic flux density is 0.25T. directed into the page. If the wire is 1.0 m long and carries 0.5A of current in the direction from the top of the page. to bottom of the page, what are the magnitude and direction of the force induced on the wire?



Solution: The direction of the force is given by the right hand rule as being to the right. The magnitude is given by,

$$F = ilB \sin\theta = (0.5 \times 1.0 \times 0.25) \sin 90^\circ$$

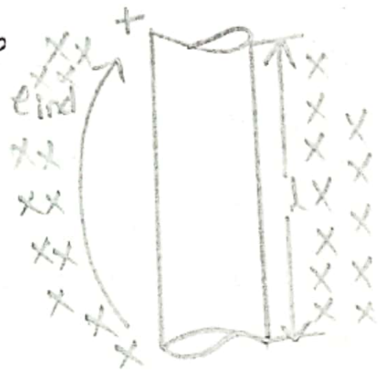
$$= 0.125N$$

Therefore, $F = 0.125N$ directed to the right.

(Ans)

1.8 A conductor moving with a velocity of $5.0\ ms^{-1}$ to the right in the presence of a magnetic field. The flux density is $0.5\,T$ into the page. and the wire is $1.0\,m$ in length, oriented as shown. What are the magnitude and polarity of the resulting induced voltage?

Solution: Since v is perpendicular to B and since $v \times B$ is parallel to l, the magnitude of the induced voltages reduces to

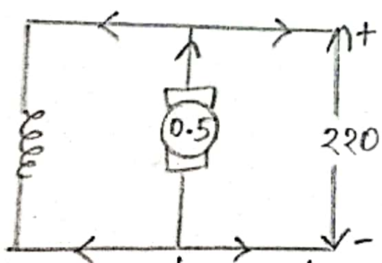


$$e_{ind} = (v \times B) \cdot l$$
$$= (vB \sin 90^\circ)\, l \cos 0^\circ$$
$$= vBl$$
$$= (5.0 \times 0.5 \times 1.0)$$
$$= 2.5\,V$$

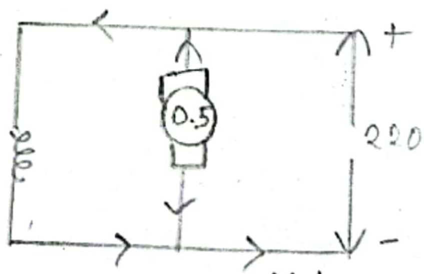
Thus the induced voltage is $2.5\,V$, positive at the top of the wire.
(Ans)

29.1 A 220V DC machine has an armature resistance of $0.5\,\Omega$. If the full load armature current is 20A. Find the induced e.m.f when the machine acts as (i) generator (ii) Motor.

Solution:



Generator



Motor

The D.C machine is assumed to be shunt connected. In each case, shunt current is considered negligible. because its value ~~its~~ is not given.

(i) As generator, $E_g = V + I_a R_a = 220 + 0.5 \times 20 = 230V$

(ii) As Motor, $E_b = V - I_a R_a = 220 - \cancel{0.5 \times}\ 0.5 \times 20 = 210V$

(Ans)

29.4 A 25-kW, 250-V dc. shunt generator has armature and field resistances of 0.06 Ω and 100 Ω respectively. Determine the total armature power developed when working (i) as a generator delivering 25 kW output and (ii) as a motor taking 25 kW input.

Solution: As Generator,

Output current $= 25000/250$

$= 100\ A$

$\therefore I_{sh} = 250/100 = 2.5\ A$

$I_a = 102.5\ A$

Generated e.m.f $= 250 + I_a R_a$

$= 250 + 102.5 \times 0.06 = 256.15\ V$

$\therefore$ Power developed in armature $= E_b I_a = \dfrac{256.15 \times 102.5}{1000}$

$= 26.25\ kW$

As motor,

Motor input current $= 100\ A$

$I_{sh} = 2.5\ A$

$I_a = 97.5\ A$

$E_b = 250 - (97.5 \times 0.06) = 250 - 5.85 = 244.15\ V$

$\therefore$ Power developed in armature $= E_b I_a$

$= 244.15 \times \dfrac{97.15}{1000}$

$= 23.8\ kW$

(Ans)